Problem Set 2
Due: Mar. 8, 2021

1. Section 1.3 Problem 16.

   We are given three coins: one has heads in both faces, the second has tails in both faces, and the third has a head in one face and a tail in the other. We choose a coin at random, toss it, and the result is heads. What is the probability that the opposite face is tails?

   Solution:

$$\begin{aligned} p &= \mathbf{P}(\text{the opposite face is tails} \mid \text{heads came up}) \\ &= \frac{\mathbf{P}(\text{the opposite face is tails and heads came up})}{\mathbf{P}(\text{heads came up})} \end{aligned}$$

$$\begin{aligned} \mathbf{P}(\text{the opposite face is tails and heads came up}) &= \mathbf{P}(\text{toss the third coin and the result is heads}) \\ &= \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6} \end{aligned}$$

$$\begin{aligned} \mathbf{P}(\text{heads came up}) &= \mathbf{P}(\text{heads came up} \mid \text{toss the first coin})\mathbf{P}(\text{toss the first coin}) \\ &+ \mathbf{P}(\text{heads came up} \mid \text{toss the second coin})\mathbf{P}(\text{toss the second coin}) \\ &+ \mathbf{P}(\text{heads came up} \mid \text{toss the third coin})\mathbf{P}(\text{toss the third coin}) \\ &= 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{1}{3} \\ &= \frac{1}{2} \end{aligned}$$

   Therefore,

$$p = \frac{1/6}{1/2} = \frac{1}{3}$$

2. Section 1.4 Problem 22.

   Each of k jars contains m white and n black balls. A ball is randomly chosen from jar 1 and transferred to jar 2, then a ball is randomly chosen from jar 2 and transferred to jar 3, etc. Finally, a ball is randomly chosen from jar k. Show that the probability that the last ball is white is the same as the probability that the first ball is white, i.e., it is $m/(m+n)$.

   Solution: We derive a recursion for the probability p_i that a white ball is chosen from the ith jar. If we choose a white ball from the ith jar, we will have $m+1$ white balls in the $(i+1)$th jar, which leads to a probability of $(m+1)/(m+n+1)$. With the same reason, If we choose a black ball from the ith jar, the probability becomes $m/(m+n+1)$. Therefore, using the total probability theorem,

$$\begin{aligned} p_{i+1} &= \frac{m+1}{m+n+1}p_i + \frac{m}{m+n+1}(1-p_i) \\ &= \frac{1}{m+n+1}p_i + \frac{m}{m+n+1} \end{aligned}$$

Starting with the initial condition $p_1 = m/(m+n)$, we have

$$p_2 = \frac{1}{m+n+1} \cdot \frac{m}{m+n} + \frac{m}{m+n+1} = \frac{m}{m+n}$$

More generally, this calculation shows that if $p_{i-1} = m/(m+n)$, then $p_i = m/(m+n)$. Thus, we obtain $p_i = m/(m+n)$ for all i.

3. A and B play a series of games. Each game is independently won by A with probability p and by B with probability $1-p$. They stop when the total number of wins of one of the players is two greater than that of the other player. The player with the greater number of total wins is declared the winner of the series.

(a). Find the probability that a total of 4 games are played.

(b). Find the probability that A is the winner of the series.

Solution:

(a). There will be 4 games if each player wins one of the first two games and then one of them wins the next two.

$$\begin{aligned} \mathbf{P}(4 \text{ games}) &= \mathbf{P}(4 \text{ games and A wins}) + \mathbf{P}(4 \text{ games and B wins}) \\ &= \mathbf{P}(\text{ABAA or BAAA}) + \mathbf{P}(\text{ABBB or BABB}) \\ &= 2p(1-p)^3 + 2p^3(1-p) \end{aligned}$$

(b). Let A be the event that A wins. Conditioning on the outcome of the first two games gives

$$\mathbf{P}(A) = \mathbf{P}(A \mid (a,a))p^2 + \mathbf{P}(A \mid (a,b))p(1-p) + \mathbf{P}(A \mid (b,a))(1-p)p + \mathbf{P}(A \mid (b,b))(1-p)^2$$

where the notation (a,b) means, for instance, that A wins the first and B wins the second game. Note that when there is a draw in the first two games, the situation becomes the same as the very beginning, i.e., $\mathbf{P}(A \mid (a,b)) = \mathbf{P}(A \mid (b,a)) = \mathbf{P}(A)$. Therefore,

$$\mathbf{P}(A) = p^2 + \mathbf{P}(A)2p(1-p)$$

Solving the above equation gives

$$\mathbf{P}(A) = \frac{p^2}{1-2p(1-p)}$$

4. Assume that each child is, independently, equally likely to be either a boy or a girl. If every family has three children, what proportion of all sons are eldest sons?

   Solution: Choose a child at random. Letting E be the event that the child is an eldest son, letting S be the event that the child is a son. The answer for the problem is $\mathbf{P}(E \mid S)$.

   $$\mathbf{P}(E \mid S) = \frac{\mathbf{P}(E \cap S)}{\mathbf{P}(S)} = 2\mathbf{P}(E)$$

   with $\mathbf{P}(E \cap S) = \mathbf{P}(E)$ and $\mathbf{P}(S) = 1/2$. Letting A be the event that the child's family has at least one son. With total probability theorem,

   $$\begin{aligned}\mathbf{P}(E \mid S) &= 2\left[\mathbf{P}(E \mid A)\mathbf{P}(A) + \mathbf{P}(E \mid A^c)\mathbf{P}(A^c)\right] \\ &= 2\left[\frac{1}{3} \cdot \left(1 - \frac{1}{8}\right) + 0 \cdot \frac{1}{8}\right] \\ &= \frac{7}{12}\end{aligned}$$

   with $\mathbf{P}(A^c) = \mathbf{P}(\text{three daughters}) = 1/8$ and $\mathbf{P}(E \mid A) = 1/3$, which represents the probability that the randomly picked child is the eldest son in a family with at least one son.